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Machine learning and IoT-based smart farming for enhancing the crop yield

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Abstract. The lack of favourable atmospheric conditions leads to the loss of many crops each year. In India alone, over 11 billion dollars are lost. By combining IoT and machine learning technologies, this team has created a system that integrates agriculture's three primary operations: crop selection, autonomous watering, and fertiliser suggestion. The following crops—Apple, Rice, Maize, Grape, Banana, Orange, Cotton, and Coffee—were considered in the study. Three systems are covered in the paper: The crop recommendation system employs machine learning to examine factors including nitrogen (N), phosphorous (P), potassium (K), pH, and weather before recommending a crop. The crop type and the current levels of soil nutrients are the two main determinants on which the fertiliser recommendation method bases its recommendation. When employing an automatic irrigation system, the crop is irrigated automatically while taking current soil moisture levels and weather forecasts into consideration. This paper attempted to implement the mentioned systems. The paper discusses the successes of the crop recommendation system, the automatic watering system, and the fertiliser recommendation system. In this paper, we report the results of simulations of the mentioned systems.

1. Introduction

With the help of a Machine learning algorithm, a system will be developed that can select crops, irrigate them autonomously, and recommend fertilizers. A system that will help farmers reduce physical labor, use less energy, and increase productivity will be developed.

The background study is the very foundation of any research. The most significant parts are discussed in this paragraph. The primary goal of this project is to develop an intelligent agriculture system to increase field yield. [1] provides great intelligence on selecting the best crop for crop rotation by routinely checking the area; it also discusses improving the irrigation system through selective irrigation. A system is developed and published in [2] that uses IoT and ML to produce an affordable smart farming module, The system uses state-of-the-art methods to improve the accuracy of the results. The scope of machine learning in precision agriculture is also explored with the help of [3], which explores machine learning in the context of food production and agriculture. This research compares the performance of machine learning in agriculture to other artificial intelligence models that have been applied in agriculture. The contrasts between several machine learning model types are also explored. The study discusses why some machine learning models are more suitable for usage in agriculture than others [4], discusses the implementation of several machine learning algorithms in sensor data analytics within

the agricultural ecosystem, and is thoroughly reviewed in this study. It also provides a case study of integrated food, energy, and water systems using an IoT-based smart farm prototype. Also, [5] provides a deep reinforcement learning-based smart agriculture IoT system with four layers: agricultural data collection, edge computing, agricultural data transmission, and cloud computing. The method mixes agricultural production with some cutting-edge information technologies, most notably cloud computing and artificial intelligence, to enhance food output. Deep reinforcement learning, the most powerful AI model, is implemented in the cloud layer to make instantaneous smart decisions such as determining the amount of water needed to be irrigated for an increasing crop growth environment. Now coming to the algorithms part of the study [6], it discusses crop recommendation using parameters like N, P, and K values and with the help of ML algorithms. A decision tree algorithm is proposed in [7] for a smart irrigation system with temperature, humidity, and moisture as parameters. [8] undertakes an extensive analysis of several machine learning concepts for an IoT-based smart agriculture system. The support vector machine (SVM) and decision tree algorithms are proposed in [9] to get the crop suitable for the given soil data and help enhance the growth using an optimised farming process. In [10], ML with computer vision is reviewed for the classification of a different set of crop images in order to monitor crop quality and yield assessment. Now coming to the systems and solutions proposed part of the study, a system is proposed in [11] for developing and testing a smart farming system built on a platform with intelligence that allows for prediction utilising artificial intelligence (AI) methods. The system's implementation entails three primary phases: data collection using sensors placed in an agricultural field, data cleaning and storage, and third, predictive processing utilising some AI techniques. The system is built on wireless sensor network technology. A solution architecture is proposed in [12] for designing key performance indicators (KPIs) in precision agriculture (PA). An IoT-based smart farming system and an effective prediction approach dubbed WPART (Wrapper feature in PART classification technique) based on machine learning techniques to forecast crop productivity and drought for effective decision-support in IoT-based smart farming systems were recommended in [13]. An intelligent irrigation system for precision farming using the Internet of Things (IoT) is proposed in [14]. The technology, which incorporates feedback, maintains its performance better in any region's weather for any length of time. To forecast the volumetric soil moisture content, irrigation schedule, and spatial distribution of water needed to irrigate the arable land, the algorithm uses a long short-term memory network (LSTM). [15] gives an overall idea about crop prediction and crop monitoring techniques using several research papers and published systems.
Using the aforementioned publications as a guide, a system is developed that can suggest crops based on weather, soil characteristics, crop requirements, and climatic circumstances and automatically irrigate crops based on crop, weather forecasts from the open weather API, and fertiliser recommendations based on crop and field characteristics.
The foremost contributions of this paper are:

- Crop Recommendation System
- Automatic Watering System
- Fertilizer Recommendation System

The organization of the paper is mentioned below:
There are four sections: an overview of the problem, potential solutions, and the solution proposed in the introduction section; a description of the proposed methodology; an overview of the three subsystems—crop recommendation, automatic watering, and fertiliser recommendation; results and discussions, which present and discuss the findings of the simulated systems; and a summary of the work that has been done.

2. Proposed Methodology

To improve crop yield, a final system that is capable of performing several functions needs to be created. For this, three subsystems are combined into one primary system, as illustrated in figure 1.

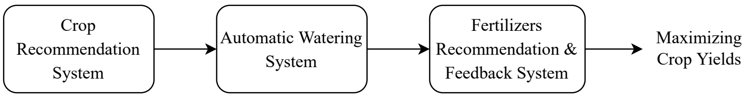

Figure 1: Proposed Methodology

The first method, known as the "crop recommendation system," examines numerous soil and meteorological conditions in a specific place and recommends a crop that would do well there. The automatic watering system, the second system, examines the soil's moisture content and the weather. Based on the plant's requirements for moisture, the system then determines if it is necessary to water the plant. The third system, known as the fertiliser recommendation system, makes suggestions about how much and when fertiliser should be applied to the crop to ensure healthy growth. Combined into one main system, these three subsystems can improve crop growth.

2.1. Crop Recommendation System

This system checks the temperature, humidity, and rainfall of the chosen location before checking the NPK and pH of the soil. While the pH data is gathered using a pH sensor, the NPK values are gathered using an NPK sensor. Then, using a cloud platform open weather api, the values of temperature, humidity, and rainfall are gathered. To suggest the optimum crops, the collected values are then run through a machine learning model, as illustrated in figure 2. The diverse data values are first clustered using K-Means clustering. The accuracy of 5 distinct machine learning models—RFC, GBC, DT, XGB, and KNN—would then be compared. Values gathered from the sensor and cloud will then be transmitted via whichever one is most appropriate. The best crop that could grow in that place would then be suggested by this system.

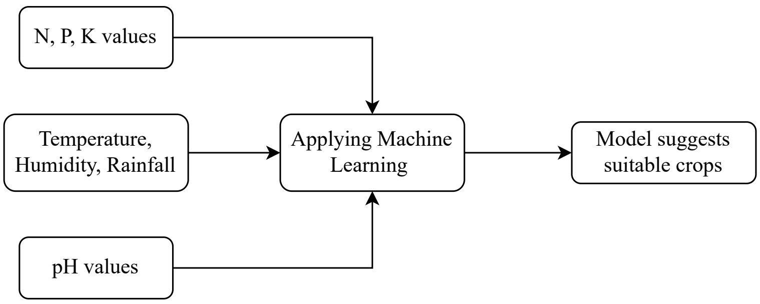

Figure 2: Crop Recommendation System

2.2. Automatic watering system

The soil moisture content will be examined using a moisture sensor, and the weather will be checked using a cloud platform open weather api. The amount of water being poured into the soil will be calculated using flow sensors. The working of the system is illustrated using figure 3. After every P hour, this system is intended to monitor the temperature values. The mechanism examines the temperature if the soil moisture level is low and rain is not imminent. If the temperature is normal, the system turns on the pump to water the plants for X minutes

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until the crop's threshold moisture need is met. If the temperature is high, the mechanism activates the pump to water the plants for X+Y minutes until the crop's maximum moisture demand is met. And if rain is predicted, the mechanism will hold off until it starts. Based on the field's size, the flow sensor values, and the amount of water the plants need, the appropriate time for watering them is determined. Microcontrollers will be used to implement this capability. The plants will receive the right amount of moisture and grow healthily thanks to this technique.

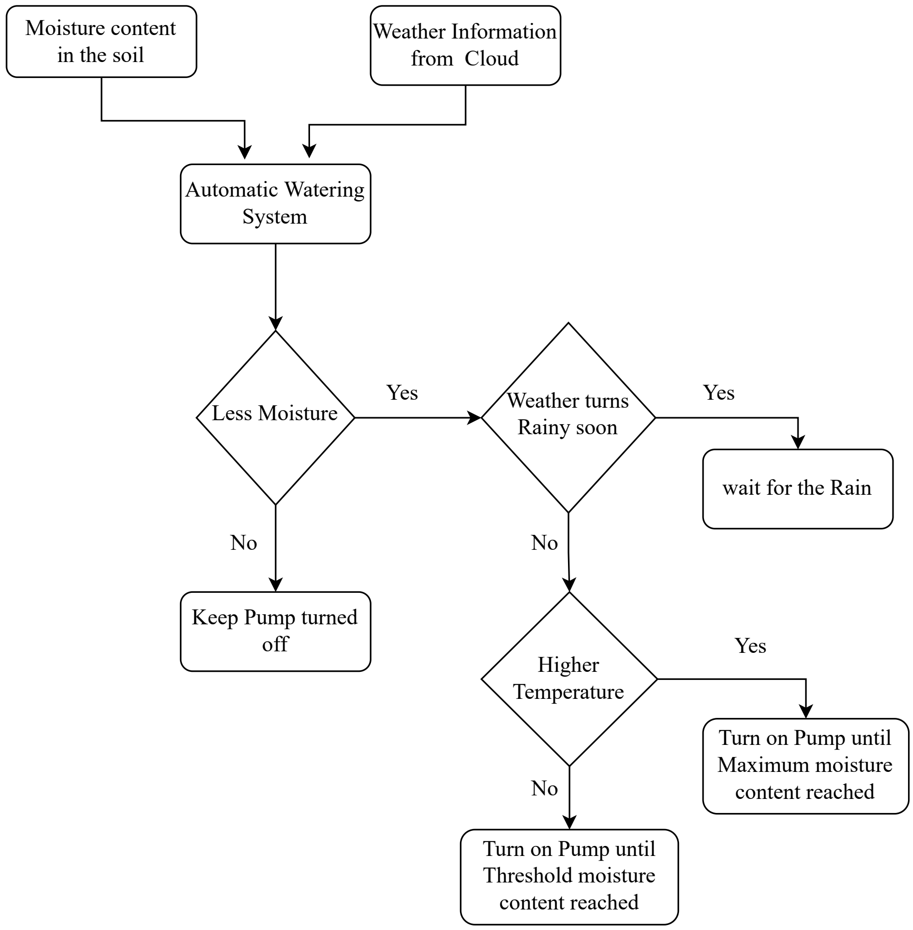

Figure 3: Automatic Watering System

2.3. Fertilizer Recommendation system

Soil fertility plays an important role in crop growth.The N, P, and K values of the soil have a significant role in determining soil fertility. An NPK sensor can be used to analyse these values. There would be a set of uniform NPK values for every crop.All the values considered were used to analyse the land where crop is being cultivated. The gathered information is analysed together with the crop being grown, as seen in figure 4. Finally, the system would determine whether fertility is at its peak. If it is ideal, normal fertilisers that would keep the soil fertile are advised. If not, additional fertilisers are advised in addition to the regular fertilisers as needed by the crop. The system also offers fertiliser dosing recommendations. This would support preserving the crop's soil fertility until it had been harvested.

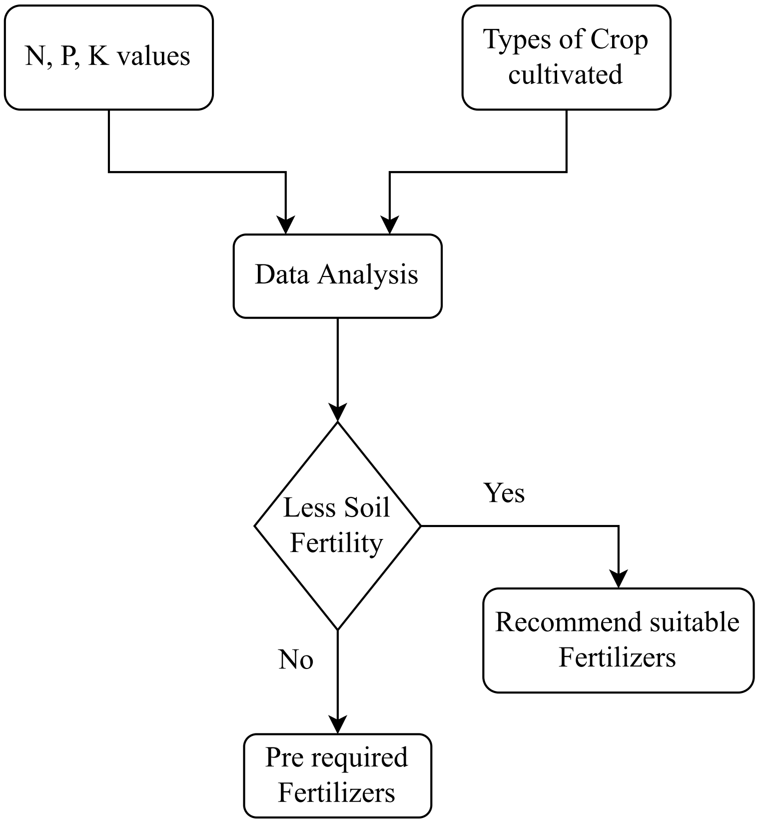

Figure 4: Fertilizer Recommendation System

3. Results & Discussion
The suggested models have been simulated, and the section contains the findings.

3.1. Crop Recommendation System

Table 1: Accuracy of various classifiers

	Model	Accuracy	Train acc
0	KNN	0.977273	0.988636
1	DT	0.993182	1.000000
2	RFC	0.997727	1.000000
3	GBC	0.995455	1.000000
4	XGB	0.993182	1.000000

In table 1, the accuracy of 5 distinct models has been compared. From the findings, KNN fits the dataset used; therefore, the KNN machine learning algorithm is implemented in the crop recommendation system.

```
Enter the value of Nitrogen:  32
Enter the value of Phosphorus:  32
Enter the value of Potassium:  2
Enter the value of pH:  2.323
Enter your City:  Noida
[[ 32.         32.          2.          30.          65.
    2.323      108.14157296]]
The Suggested Crop for Given Climatic Condition is : Orange
```

Figure 5: Crop Recommendation System simulation Results

The pH value, as well as the values for N, P, and K, are shown as input in figure 5. The humidity, rainfall, and temperature values are then requested from the cloud for the city provided as input. To determine the best crop that can be cultivated in the area, all these values are then run through the KNN machine learning model. Based on the values provided, the crop suggested for cultivation is an orange.

Table 2: Crops recommended for different atmospheric conditions

N	P	K	Temperature	Humidity	pH	Rainfall	Crop Reco.
90	42	43	20.87974371	82.00274423	6.502985292	202.9355362	Rice
8	120	201	21.18667419	91.13435689	6.321152192	122.233323	Apple
71	54	16	22.61359953	63.69070564	5.749914421	87.75953857	Maize
13	144	204	30.7280404	82.42614055	6.092241627	68.38135469	Grape
117	76	47	25.56202173	77.38229006	6.119216009	93.10247183	Banana
12	29	13	22.45616931	91.52781832	7.57125447	118.0069295	orange
131	35	18	24.49112609	82.24415809	7.057693366	64.02949379	Cotton
84	39	35	23.17714381	52.13864034	6.959404135	117.3113562	Coffee

The crop recommendation system's recommendations for various sets of input data are shown in table 2. This demonstrates how different crops are best adapted to grow under various atmospheric conditions.

3.2. Automatic Watering System

Table 3: Daily Water requirement of various crops

Crop	Minimum(in mm)	Maximum(in mm)
Apple	930	1080
Rice	1300	1500
Maize	400	600
Grape	600	730
Banana	900	1200
Orange	850	1100
Cotton	500	650
Coffee	1100	1500

Table 3 displays the range of water requirements for each crop in mm. For a given crop, it would promote healthy growth if the soil moisture content fell within this range. The system would ensure that the crop's soil moisture was maintained.

```
Enter your City:  karaikal
27
Enter Current Crop:  orange
Enter the current moisture level:  5
Water supply is required
Supply of Water Required for 7.5s
```

Figure 6: Automatic Watering System simulation Results

The cultivated crop is input along with the city where it is grown, as shown in figure 6, to gather data on temperature and rainfall from the cloud. Farmland of $1m^2$ in size is assumed, with 1 mm of water shortage filled for every 1 s of water flow past the flow sensor. It can be seen from the input data that 7.5 seconds of total pump time are needed to meet the water demands.

3.3. Fertilizer Recommendation System

Table 4: NPK requirements of different crops

Crop	N	P	K
Apple	20.8	134.2	199.89
Rice	79.89	47.58	39.87
Maize	77.76	48.84	19.79
Grape	23.18	132.53	200.11
Banana	100.23	82.01	50.05
Orange	19.58	16.55	10.01
Cotton	117.77	46.24	19.56
Coffee	101.24	28.74	29.94

The NPK values are crucial for achieving optimal crop growth outcomes.Table 4 displays each crop's N, P, and K needs. The system looks for these numbers and proposes appropriate fertilisers to be utilised for the crop's healthy growth depending on those results.

```
Enter a crop:  grape

    Urea = 21.73 Kg/Hec
    MOP = 16.66 Kg/Hec
    SSP = 62.5 Kg/Hec

    Dosage Recommendation = Once every month
```

Figure 7: Fertilizer Recommendation System simulation Results

The N, P, and K readings from the soil collected by the NPK sensor, as well as the crop being grown, were provided as input. Following an analysis of these inputs, fertilisers and dosage are recommended, as shown in figure 7. This would enable monitoring changes in soil NPK values.

4. Conclusion

A system has been developed that can recommend crops based on the local climate and soil characteristics, water plants automatically when necessary, and recommend the optimum fertilisers based on the crop type. The technique can be applied to the following crops for the best results: apple, rice, maize, grape, banana, orange, cotton, and coffee. This technology could be used by farmers as an effective decision-making tool to help them reduce manual labor, conserve energy, and increase output. For the future scope of this work, a disease prediction system can be added to the suggested system to predict crop diseases based on image classification.

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